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A Survey on AI-Based Detection Methods of GPS Spoofing Attacks on UAVs

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***Abstract*— Global Positioning System (GPS) spoofing attacks on unmanned aerial vehicles (UAVs) can cause severe consequences, including abnormal behavior, privacy breaches, safety hazards, financial losses, and national security risks. Hence, detecting and preventing GPS spoofing attacks on UAVs is paramount to ensure their safety and security. This survey paper explores the cutting-edge techniques based on artificial intelligence (AI) proposed in the literature for detecting GPS spoofing attacks on UAVs. We present an overview of the UAV navigation system and then delve into the concept of GPS spoofing, encompassing its various forms, such as soft and hard spoofing. Furthermore, we discuss the advantages of utilizing AI in developing detection methods compared to traditional approaches. Subsequently, we analyze the detection methods based on deep learning (DL) and machine learning (ML) proposed in the literature, evaluating their strengths and weaknesses whenever feasible. Our survey provides valuable insights for researchers and practitioners working in the fields of AI and UAV security with the target of developing more advanced and effective AI-based methods for GPS spoofing detection on UAVs.**

Keywords- drone; deep learning; machine learning; detection; GPS spoofing; unmanned aerial vehicle; UAV

NOMENCLATURE

AI	Artificial Intelligence
ANN	Artificial Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
BOX-MLP	BOX Multi-Layer Perceptron
CART	Classification and Regression Trees
CBA	CNN-BiLSTM-Attention
CNN	Convolutional Neural Network
DL	Deep Learning
DNN	Deep Neural Network
DoS	Denial of Service
GAN	Generative Adversarial Network
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GRU-RNN	Gated Recurrent Unit Recurrent Neural Network
IDS	Intrusion Detection System
IF	Isolation Forest
IMU	Inertial Measurement Unit
INS	Inertial Navigation System
KNN	K-Nearest Neighbor
LIDAR	Light Detection and Ranging
LOF	Local Outlier Factor

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), often called drones, are aircraft that operate without a human pilot on board. A human operator can control those vehicles remotely or navigate independently based on predetermined flight paths. UAVs come in various sizes, ranging from small handheld models to large-scale aircraft. They are employed for various applications and can be equipped with different sensors and imaging devices to collect aerial data and capture photographs. Additionally, certain UAVs are specifically designed to transport specific payloads, such as weapons, supplies, or other types of equipment.

In recent years, the use of UAVs has experienced significant growth, and they are gaining popularity across various industries. They can be used in a wide range of applications: Aerial photography and videography. UAVs equipped with high-quality cameras capture aerial images and videos for various purposes, such as cinematography, real estate, construction, surveying, and agriculture. Also, UAVs equipped with multispectral and hyperspectral sensors monitor crops and soil health, enabling farmers to make more informed decisions about their fields. In the same context, search and rescue UAVs equipped with thermal imaging and other sensors are used to locate missing persons and identify hazards in disaster zones, providing valuable information to rescue teams. Similarly, delivery services UAVs are being developed and tested for package delivery, especially in hard-to-reach areas, inspection, and maintenance. In this area, UAVs are used to inspect infrastructure such as power lines, bridges, and pipelines for damage and maintenance, reducing the need for human inspectors to work at heights or in dangerous environments.

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Nowadays, for security and surveillance, UAVs are used by law enforcement and military organizations for surveillance and monitoring of significant events, crowd control, and border surveillance; environmental monitoring, UAVs are used for environmental monitoring and conservation efforts, such as tracking wildlife populations and monitoring deforestation; mapping and surveying, UAVs equipped with Light Detection and Ranging (LIDAR) and photogrammetry sensors are used for mapping and surveying land areas for various purposes, such as urban planning and construction [1]. The above examples represent only a glimpse of what UAVs could accomplish. With ongoing technological advancements, it is expected that novel uses will emerge. Nonetheless, before UAVs can be broadly implemented, several hurdles exist, including ethical, privacy, and safety concerns. UAVs are also vulnerable to many cybersecurity attacks [2].

Like all network-connected devices, UAVs are susceptible to cybersecurity attacks that compromise their safety and functionality. Here are some of the most common cybersecurity issues in UAVs: data security: UAVs may gather large amounts of data, which can be sensitive or confidential; malware and viruses, UAVs can be infected with malware or viruses, which can cause the UAV to malfunction or perform unintended actions; denial of service (DoS) attacks, attackers can overload UAV communication channels with data, causing the UAV to crash or lose connection with its operator; rogue UAVs, attackers can use UAVs as weapons or as reconnaissance tools to gather intelligence on targets; GPS spoofing, GPS is a critical component of UAV navigation, and attackers can manipulate the GPS signal to misdirect or even hijack the UAV and potentially cause harm. The latter attack is the primary concern of this paper. Most of the applications, as mentioned earlier, lie on the location of the UAV to provide the intended service. Therefore, it is of high importance to detect and mitigate such attacks. GPS spoofing attacks have been addressed intensively, in general, in autonomous vehicles and other applications. Some recent studies can be found in [3] – [4].

In this paper, we review and discuss some of the state-of-the-art works in the literature that introduced methods to detect GPS spoofing attacks, specifically on UAVs. We explain the UAV navigation system and the types of GPS spoofing attacks that may occur on such systems. We also describe the differences between the AI-based and traditional-based detection approaches. In this paper, we mainly concentrate on the works that proposed AI-based methods since they use intelligent techniques to learn complex patterns in data and consequently may enhance the accuracy of GPS spoofing attack detection in complex scenarios. We aim to be both comprehensive and inclusive in this survey. We provide a comparative analysis to highlight the pros and cons of each proposed method in the literature. It is worth mentioning that we assume proper knowledge of the reader of ML and DL as we do not explain the algorithms or neural networks themselves in this paper but concentrate on their use in detecting spoofing attacks on UAVs. We aim from this work to provide valuable insights for researchers and practitioners working in the fields of AI and UAV security with the target of developing more advanced and effective AI-based methods for GPS spoofing detection on UAVs due to its widespread in many essential and critical applications as have been described before.

The rest of the paper is organized as follows. The UAV cybersecurity threat model is discussed in Section II. The functionality of the UAV navigation system and the GPS spoofing attacks on UAVs are discussed in Section III. The advantages of adopting AI algorithms in developing methods to detect GPS spoofing attacks on UAVs over the rest are discussed in Section IV. State-of-the-art ML-based and DL-based detection methods proposed in the literature are addressed in Sections V and VI, respectively. The findings are discussed in Section VII, and the paper is concluded in Section VIII.

II. UAV CYBERSECURITY THREAT MODELS

It is imperative to briefly discuss the general cybersecurity threat model applicable to UAVs and identify potential vulnerabilities. More elaborations on cybersecurity threat models can be found in many works in the literature, such as [5] – [8].

Cybersecurity threat models for UAVs often categorize risks based on the layers of UAV architecture, communication, control systems, and payload systems. Each of these layers presents distinct vulnerabilities that can be exploited by adversaries. For instance, communication links between UAVs and ground control stations are particularly vulnerable to eavesdropping, signal spoofing, and jamming attacks. Researchers emphasize that intercepting or disrupting communication can lead to loss of operational control or misdirection of the UAV. The control systems of UAVs, including flight and mission control algorithms, are susceptible to attacks aimed at the software level. These systems rely heavily on the integrity of the data they receive, which could result in incorrect commands and potentially catastrophic outcomes if corrupted through malware or other forms of cyber attacks. Additionally, the payload systems, which may include cameras, sensors, or other data-collecting devices, can be compromised. This could lead to espionage, data theft, or manipulation of data, which is a significant threat in military and surveillance applications. Another potential vulnerability arises from the physical interfaces and maintenance ports on UAVs. Unauthorized access to these can allow attackers to bypass network security measures and directly manipulate UAV hardware and software.

To effectively address and reduce the cybersecurity risks associated with UAVs, it is crucial to implement a multi-layered security approach as advised by current threat models. This comprehensive strategy involves the deployment of robust, end-to-end encryption for all communication channels. Encryption serves as the first line of defense for protecting the integrity and confidentiality of data transmitted between UAVs and ground control stations. Implementing robust encryption protocols such as AES-256 shields UAV communications from eavesdropping and spoofing attacks. Additionally, conducting continuous and rigorous software testing is crucial to address vulnerabilities within UAV control systems. This includes regular updates and patches to fix any

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identified security gaps. Moreover, UAV software should undergo extensive testing phases, including penetration testing and vulnerability assessments, before deployment. These tests help to simulate potential attacks and identify weak points within the system's software architecture. Moreover, physical security measures are as crucial as cyber defenses in protecting UAV systems. Secure access controls for physical interfaces prevent unauthorized access to UAV hardware. This involves the use of tamper-proof hardware modules, where critical components are sealed and shielded against physical tampering. Additionally, using biometric data, such as fingerprint recognition or retina scans, ensures that only authorized personnel have access to UAV maintenance ports and internal systems. Furthermore, the integration of advanced anomaly detection systems is essential; these systems monitor operations to detect any irregular activities that could indicate a security breach, providing the opportunity for quick defensive actions to mitigate potential threats. By monitoring network traffic and system behavior, these systems can detect deviations from normal operations, which may signify an ongoing attack or system compromise. AI algorithms enhance the sensitivity and accuracy of anomaly detection, enabling these systems to adapt and respond to new and sophisticated threats dynamically. Early detection through these systems allows for rapid response strategies, minimizing potential damage and reinforcing the UAV's resilience against attacks. This multi-faceted approach is designed to enhance UAV operations' overall security and resilience. By integrating these security measures, UAV operators and manufacturers can fortify their systems against various cyber threats.

In this paper, we emphasize one category of attacks, which are the GPS spoofing attacks on UAVs and their AI-based detection methods. We describe such attacks in the following section.

III. GPS Spoofing Attack on UAVs

Navigation systems on UAVs are critical for ensuring that the UAVs can fly safely and accurately. UAVs rely on either one of the following devices or a combination of them: GPS receivers, onboard motion sensors, and vision sensors to navigate through the airspace and reach their intended destinations. GPS positioning in the states is based on signals with codes sent by 24 satellites, which are located 20,192 km above the Earth and collected by GPS receivers. The orbital range of these 24 satellites around the Earth ensures that a GPS receiver is visible from at least four satellites anytime and anywhere [9]. A GPS receiver processes the codes received, from the satellites, which provide information about the time when the signal was transmitted and the location of the satellites. A mathematical method called trilateration is used to obtain the location of the receiver related to the intersection of three spherical surfaces, which represents the distances between a GPS receiver and three of the 24 satellites mentioned earlier. One of the most critical factors for the accuracy of GPS is the synchronization between the clock in the GPS receiver and the highly accurate atomic clocks in the GPS satellites. The signals that satellites send include the precise time at which the signal was transmitted. The receiver determines its distance from each satellite by calculating the

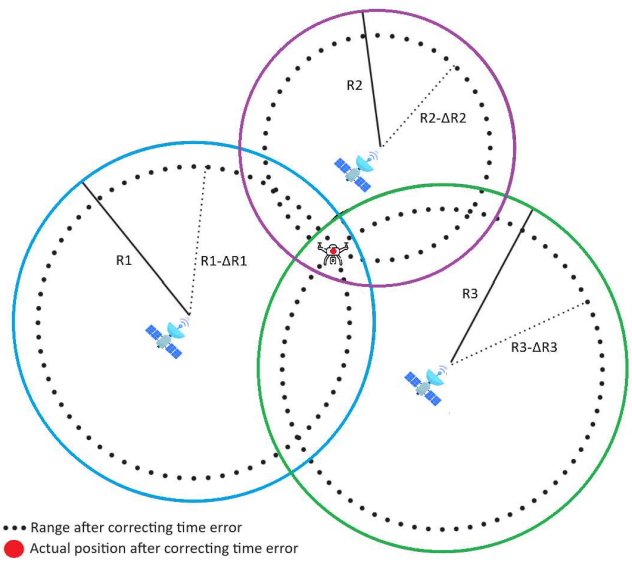

Figure 1. GPS trilateration method to locate objects. The different Rs (R1, R2, and R3) represent the radiuses of the coverage without time error corrections, while the Rs-ΔRs (R_1-ΔR_1, R_2-ΔR_2, R_3-ΔR_3) represent the radiuses of the coverage with time error corrections.

time delay of the incoming signal. However, GPS receivers typically do not have atomic clocks due to their size and cost. This leads to errors in distance measurements due to slight inaccuracies in the receiver's clock. By using signals from at least four satellites, the GPS receiver can solve for its spatial coordinates in addition to the time error in its clock. Correcting this time error significantly improves the accuracy of the position calculation. This is illustrated in Fig. 1. Satellite-based GPS navigation may provide reasonably stable accuracy if enough GPS signals are tracked over the whole UAV mission. One drawback of using GPS alone in UAVs is its inability to provide accurate altitude measurements, which is essential for UAVs [10]. Hence, GPS functionality is enhanced through the integration with an Inertial Navigation System (INS). Basically, the system works by fusing sensory information taken from inertial sensors such as accelerometers and rotational sensors such as gyroscopes to continuously estimate the position and orientation of the UAV in real-time using computer algorithms. An INS needs to work on a significant growth in systematic errors when used alone due to the integration of different sources of sensor errors over time [11]. Utilizing an INS accompanied by a GPS receiver, the system provides position, velocity, and altitude navigation parameters with high data rates and good accuracy. Although the use of a GPS/INS solves the previously mentioned problems of using a GPS receiver or an INS separately, the integrated system fails to provide crucial information about the vertical distance and movement of a UAV relative to the ground, as stated in [10]. The use of a vision-based navigation system, which relies on vision sensors such as cameras, hyper-spectral sensors, laser scanners, etc., integrated with the GPS/INS system could encounter the problem mentioned earlier. Vision-based navigation is mainly used to process a sequence of stereo imagery to determine the platform trajectory, which, therefore, solves the issue of the poorly determined platform trajectory by the GPS/INS system [10].

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The primary component of the integrated navigation system responsible for directing UAVs in their intended direction is the GPS. However, GPS spoofing attacks, which generate counterfeit GPS signals, can manipulate the UAV's navigation system and consequently cause abnormalities in the UAV missions and behavior. The possible spoofing attack to mislead the GPS receiver is elucidated in the following.

GPS spoofing attacks can be classified into two types, namely, soft and hard GPS spoofing, as discussed in [12]. Soft GPS spoofing involves transmitting a counterfeit signal at the same frequencies as the satellites involved in the positioning of the UAV. With some degree of synchronization and a transmitted power slightly higher than the power of the authentic GPS signal to match the authentic signal's power level, the GPS receiver is spoofed. In other words, for a soft GPS spoofing to be successful, the counterfeit signal must meet the following requirements: having the same power level as the authentic signal, transmitted at a distance within the antenna coverage of the threatened GPS receiver, and the delay time is analogous to the ones of the authentic signal. A hard GPS spoofing attack occurs when one or more of the requirements for soft GPS spoofing are not met. In this case, a jamming signal is created and transmitted at high power to disrupt the authentic signal. The jamming signal is transmitted at the same frequency as the authentic signal, causing the GPS receiver to lose the authentic signal and lock onto the jamming signal.

The authors of [13] identify two types of GPS attacks with different levels of sophistication. Unauthenticated and authenticated GPS attacks. The former is typically carried out on civilians. Such attacks use unauthenticated signals, and hence, the attacker can create spoofing signals, change the content of the authentic GPS signals, and/or delay signals or send them ahead of time. On the other hand, authenticated GPS attacks target military UAVs that require sophisticated and expensive equipment to perform the attack. Attackers cannot produce military GPS signals, as they are authenticated, but can catch and relay signals, which may delay but not send signals ahead of time. According to the categories mentioned in [12], the authors of [14] considered the unauthenticated GPS attack as soft and the authenticated GPS attack as hard.

GPS spoofing attacks on UAVs can manifest in a multitude of ways, each with its unique repercussions depending on the context. Key impacts include unpredictable UAV behavior, invasion of privacy, compromised data integrity, safety concerns, critical mission disruption, financial implications, and national security threats. The extent of the impact of GPS spoofing attacks varies based on factors such as the UAV's type, its application, and the attack's context. While a minor deviation might be inconsequential in some missions, it can lead to critical outcomes in others, like military operations. Therefore, recognizing and mitigating such attacks is imperative for the operational integrity and security of UAVs and their applications. An illustration of a GPS spoofing attack is shown in Fig. 2.

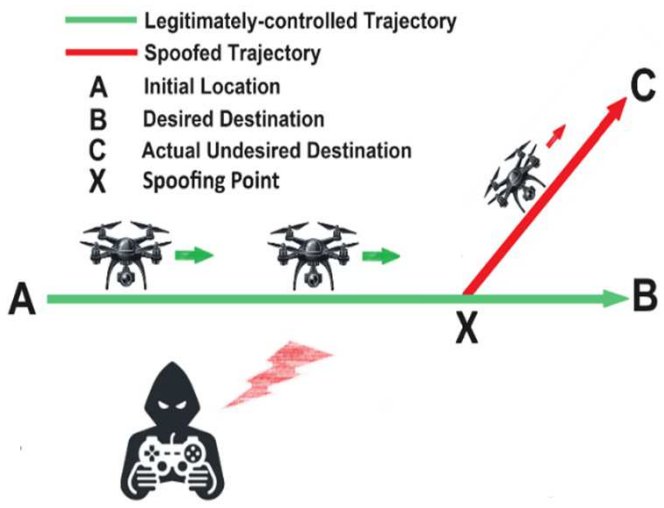

Figure 2. GPS spoofing attack illustration.

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An illustration of a GPS spoofing attack is shown in Fig. 2. The path AXB is the original predetermined trajectory or the desired trajectory, which is determined in real-time using a remote controller; on the other hand, the path AXC is the spoofed trajectory. At point X, the attacker performed a successful spoofing attack, where the UAV changed its direction toward the undesired destination.

IV. AI-Based vs. Traditional-Based Detection Approaches

Recently, many techniques have been adopted to detect anomaly behaviors of UAVs, in general, using intrusion detection systems (IDS). Some of them are traditional-based approaches, such as the works in [15] – [17], while others are AI-based approaches, such as the works in [18] – [46]. An IDS using AI can be called a learning-based approach and does not require the theoretical creation of an accurate formal system; this works by studying historical data to understand the system's behavior. Attacks on UAVs are random and unexpected; this makes the learning-based approach a fast, timely, and robust detection method. Moreover, when an attack happens, the outcome will significantly differ from the actual standard data. In that case, the AI IDS will allow the avoidance of attacks by promptly transmitting decision information to the UAV microprocessor. Nevertheless, IDS trained through AI can detect defects at the hardware level and software levels. The AI-based approaches can be categorized into ML and DL. ML is a broad domain that encompasses various algorithms and techniques for enabling computers to learn from and make predictions or decisions based on data. These methods include, but are not limited to, logistic

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regression, decision trees, support vector machines, random forests, etc. ML detection methods rely on extracting handcrafted features from data, which are then used to train a model capable of making predictions. Deep learning, on the other hand, refers specifically to a class of machine learning that involves neural networks with many layers. DL techniques automate the feature extraction process, allowing these models to learn high-level abstractions in data directly from raw input. Therefore, while categorizing AI-based detection methods into ML and DL categories is technically accurate, it is essential to communicate that DL is a specialized, more advanced branch of ML. This distinction highlights the evolution from algorithms that require manual feature selection to more sophisticated models that learn to identify relevant features autonomously. The use of DL-based IDSs generally outperforms ML-based IDSs since the former uses intelligent techniques with feature extraction processes to learn complex patterns in data. Consequently, it may enhance the accuracy of GPS spoofing attack detection. However, untrained enough IDS can lead to higher percentages of missed detections or false positives, which can be avoided by proper training on large datasets. Also, a high computational load is placed on the UAV, which is not actually a drawback nowadays with the rapid advancements of powerful microcontrollers.

Traditional approaches have been used to detect anomaly behaviors in UAVs, including GPS spoofing attacks, but they have been inefficient and require regular updating processes. The traditional-based approaches can be categorized into two types, namely, redundancy-based approaches and behavioral-based approaches, as mentioned in [18]. The following discusses their limitations in the use of GPS spoofing attacks and, hence, the superiority of the AI-based approaches. Redundancy-based approaches [15], [16] often use extra hardware components to perform cross-checks and cross-correlation of their status during runtime. For example, a dual receiver can be used in some cases to enable cross-correlation between the signals in the receivers, which also requires an additional communication link between the receivers. The redundancy-based approaches invariably add to the cost and complexity of an IDS. On the other hand, behavioral-based approaches [17] require a description of how a normal UAV operates implemented at the program level. If the UAV behaves differently, then an attack may occur without detection. A precise description of the UAV behavior is hard to predict as its behavior is generally stochastic and depends on the surrounding environment, so false positive detection is possible with high rates.

V. ML-Based Detection Methods

In this section, we discuss the various ML-based detection methods of GPS spoofing attacks introduced in the literature, specifically for UAVs. We discuss each of them in the following. We also summarize their findings in Table I.

Jasim et al. proposed in [20] an ML-based system for UAVs that harnessed ML techniques to detect jamming and spoofing attacks. They utilized a UAV Attack Dataset [19], comprising various features including timestamp, valid, timestamp sample, thrust, hover_thrust_var, accel_innov, accel_innov_var, accel_innov_test_ratio, accel_noise_var (inputs of classification phase), and label (output of classification phase). These features were utilized to train and test their system. However, they conducted training and testing using an 80% and 20% split, respectively, which showed that it was the most effective approach compared to other splits. The authors proposed several classifiers, including K-Nearest Neighbor (KNN), Gradient Boosting, and Decision Tree, and evaluated their performance using different metrics. Among these classifiers, the Decision Tree model achieved outstanding results. The proposed Decision Tree exhibited an accuracy rate of 99.86% while achieving high recall, F1-score, and precision values of 99.93%, 99.92%, and 99.93%, respectively. With a low error rate of 0.14%, the Decision Tree model effectively detected spoofing attacks. Furthermore, the proposed Decision Tree showed efficient execution with a duration of 0.66 seconds. Future work may consider the utilization of alternative classifiers, such as random forest (RF) or support vector machine (SVM), to enhance system performance further.

Manesh et al. focused in [21] on detecting GPS spoofing signals in UAVs using a supervised ML approach based on an artificial neural network (ANN). The authors presented their created dataset and investigated the effectiveness of different features, including pseudo-range, Doppler shift, and signal-to-noise ratio (SNR), in accurately classifying GPS signals. They extensively analyzed five selected features: satellite number, carrier phase, pseudo-range, Doppler shift, and SNR. When comparing the efficiency of different features for detecting GPS spoofing signals, the authors employed a neural network architecture consisting of one input layer, one hidden layer with ten hidden nodes, and one output layer. Notably, when combining the features of the pseudo-range, Doppler shift, and SNR with other features, an accuracy of at least 98% is achieved. The work also examined the impact of the number of hidden neurons in one-hidden-layer and two-hidden-layer neural networks on the detection performance. Increasing the number of hidden neurons from 2 to 10 in the one-hidden-layer network improves accuracy by 9.2%. However, after approximately 14 hidden neurons, the accuracy plateaus as the network becomes saturated. In contrast, the two-hidden-layer network achieved comparable accuracy with only six neurons (3 neurons in each hidden layer). The authors suggested future research directions, including exploring improved ML techniques, incorporating online learning in neural networks, and investigating unsupervised ML algorithms for classifying unlabeled data and clustering unknown received messages. Avgin et al. used in [22] the UAV attack dataset [19] to compare different classifiers, including Decision Tree, multilayer perceptron classifier (MLPC), KNN, and C-Support Vector. The authors employed RF as a feature extraction method. The primary objective of this research is to identify the ML algorithm that achieved the highest F-1 scores. The results revealed that the decision tree classifier exhibited the best performance with an accuracy of 88.16%, recall of 99.05%, F-1 score of 78.56%, and precision of 79.44%. The authors emphasized the importance of increasing the F-1 score as a critical evaluation metric for detecting UAV attacks.

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TABLE I. SUMMARY OF RESULTS FOR ML-BASED DETECTION METHODS OF GPS SPOOFING ATTACKS ON UAVS

Ref.	AI Method	Additional Info.	Dataset	Evaluation Metrics							
				AUC (%)	*Accuracy (%)*	*Recall (%)*	*F1-Score (%)*	*Precision (%)*	*MSE*	*Detection Time*	*Detection Rate (%)*
[20]	KNN	80% training 20% testing	UAV Attack Dataset	-	97.16	97	97.33	97		-	-
		70% training 30% testing		-	97.1	-	-	-	-	-	-
		60% training 40% training		-	96.71	-	-	-	-	-	-
	Gradient Boosting	80% training 20% testing		1.13	98.87	99	99.66	99		-	-
		70% training 30% testing		-	98.12	-	-	-	-	-	-
		60% training 40% training		-	98.49	-	-	-	-	-	-
	Decision Tree	80% training 20% testing		0.14	99.86	99.93	99.92	99.93		-	-
		70% training 30% testing		-	99.86	-	-	-	-	-	-
		60% training 40% training		-	99.74	-	-	-	-	-	-
[21]	ANN	-	Developed	-	98.30	-	-	-	-	-	99.2
[22]	Decision Tree	-	UAV Attack Dataset	-	88.17	99.05	78.56	79.44	-	-	-
	MLPC			-	67.93	98.72	37.32	51.78	-	-	-
	KNN			-	94.83	99.71	70.40	90.40	-	-	-
	C-SVM			-	94.73	99.73	67.38	90.21	-	-	-
[23]	Linear Regression	The following results are the average of K=5, 10, 15, and 20 (when K-folds are applied)	Developed	-	51.5	100	49	33	-	-	-
	SVM Polynomial			-	94.5	92	95.50	99	-	-	-
	SVM Linear			-	52.5	100	49	33	-	-	-
	SVM rbf			-	92.5	88	95	99	-	-	-
	SVM Sigmoid			-	15.5	19	22.5	32	-	-	-
	Naïve Bayes			-	88.5	74	85	99	-	-	-
	Decision Tree			-	92.5	89	89.75	91	-	-	-
	RF			-	93.5	89	89.75	95	-	-	-
[24]	MOD	-	Developed	-	99.8	-	-	-	-	-	99.9
	WMOD			-	99.8	-	-	-	-	-	99.9
	Bagging			-	99.6	-	-	-	-	-	99.6
	Boosting			-	99.56	-	-	-	-	-	99.35
	Stacking			-	99.7	-	-	-	-	-	99.8
[25]	Gaussian Naïve Bayes	-	Developed	-	≈91	-	-	-	-	-	≈86
	CART			-	≈99.8	-	-	-	-	-	≈99.9
	LR			-	≈91	-	-	-	-	-	≈86
	RF			-	≈99.8	-	-	-	-	-	≈99.9
	LSVM			-	≈95	-	-	-	-	-	≈92
	ANN			-	≈95.9	-	-	-	-	-	≈96.1
	PCA			-	96.34	-	-	-	-	-	98.85
	K-means clustering			-	86.23	-	-	-	-	-	88.1
	Autoencoder			-	99.53	-	-	-	-	-	99.73
[26]	KNN	-	Developed	-	79.08	-	-	-	-	-	70.75
	Radius Neighbor			-	73.81	-	-	-	-	-	71.97
	Linear SVM			-	64.56	-	-	-	-	-	62.08
	C-SVM			-	91.85	-	-	-	-	-	88.86
	Nu-SVM			-	92.78	-	-	-	-	-	91.26
[27]	PCA+OCC	-	Developed	-	-	-	90.57	-	-	-	-

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Shafique et al. proposed in [23] a methodology that involves deploying multiple ML algorithms, including Decision Tree, SVM with different kernels, Logistic Regression (LR), RF, and Naïve Bayes, for the proposed model. GPS signal characteristics, including jitter, jitter (absolute), jitter (local), jitter (RAP), jitter (ppq5), shimmer, shimmer (local), shimmer (dB), shimmer (apq3), shimmer (apq5), and frequency modulation, are used as features in the model to detect whether an attacker or a legitimate entity sends a signal. The authors collected their dataset specifically for this study. Following the analysis, SVM with the polynomial kernel was selected as the most suitable ML algorithm for the proposed work, providing better results than the other ML algorithms. Once the suitable ML algorithm was chosen, four different models (K-learning models) were developed using the K-fold validation method to enhance the model's performance and robustness. Soft and hard voting techniques were employed to assign class labels to unseen or test data. The extensive experiments and analyses demonstrated the proposed model's effectiveness, with 99% accuracy, 98% precision, 99% recall, and 98% F1-score. Furthermore, the paper suggests potential enhancements by incorporating DL algorithms or a combination of ML and DL algorithms, emphasizing integrating DL strategies with a CNN.

The authors Khoei et al. proposed in [24] two innovative, dynamic selection strategies, dubbed metric-optimized dynamic (MOD) and weighted-metric-optimized dynamic (WMOD). Typically, their methodology incorporated a unique one-stage ensemble feature selection approach that eliminates redundant or insignificant features, streamlining the detection process. The study conducted a comprehensive comparison of ten machine learning models, assessing their performance based on various metrics like accuracy, detection probability, false alarm probability, misdetection probability, and processing time. These dynamic techniques proved to surpass the traditional ensemble models, achieving remarkable outcomes. They achieved 99.6% accuracy, a 98.9% detection probability, a low false alarm rate of 1.56%, a misdetection rate of only 1.09%, and a swift processing time of merely 1.24 seconds. The study's significance lies in its innovative feature selection approach, dynamic selection methods, and thorough comparative analysis. The work utilized a meticulously balanced dataset of real and spoofed GPS signals, which underwent extensive preprocessing, including null value and noise management, encoding, normalization, and hyperparameter tuning using Bayesian optimization. This rigorous approach ensures the reliability and robustness of the detection methods. In other work, Khoei et al. presented in [25] a performance analysis of supervised and unsupervised ML models for detecting GPS spoofing attacks. The optimal simulation environment was set for each model, which then concluded that the highest achiever out of the supervised and unsupervised models was classification and regression trees (CART); it had the highest accuracy and probability of detection out of all the models with almost 100%, as well as the best performance evaluation with the lowest processing time of 1.25s, training time of 1.14s, prediction time of 0.11s, and the smallest memory size of 142.6MiB.

Aissou et al. proposed in [26] a detection method using instance-based learning models. Five models were evaluated, namely KNN, Radius Neighbor, Linear SVM, C-SVM, and Nu-SVM. Satellite signals were collected, and features were extracted using software-defined radio units. Simulations were performed on three types of GPS spoofing attacks (simplistic, intermediate, and sophisticated), and the performance of the models was compared. The results show that Nu-SVM outperforms the other models with 92.78% accuracy, 91.26% probability of detection, 6.02% probability of false alarm, and 8.73% probability of misdetection. The model also demonstrated strong computational efficiency regarding memory usage and processing time. The authors discussed the limitations of existing techniques and highlighted the contributions of their work, including real-time feature extraction, attack simulation, feature selection, and performance investigation of the models.

Whelan et al. proposed in [27] an approach that utilizes PCA and OCCs, allowing the use of flight logs for training data. The IDS, called MAVIDS, operates onboard the UAV within a resource-constrained agent device, enabling detection and potential mitigation even when communication to the ground control station is lost due to jamming. The authors discussed results focusing on GPS spoofing and jamming attacks using the F1-score as the primary metric, scoring 90.57% against GPS spoofing attacks.

Semanjski et al. proposed in [28] a method that utilizes cross-correlation monitoring and SVM classification. The data used for training and validation include synthetic and real-world datasets comprising measurements from GNSS receivers. Correlation analysis and principal component analysis (PCA) are employed for variable selection and dimensionality reduction. The findings show the efficiency of the approach in detecting signal tampering and the uniqueness in the field of cyber threat analytics compared to the existing literature on the use of SVM-based approaches to detect GNSS signal tampering attempts. Furthermore, the authors highlighted the importance of prioritizing precise detection over mere success rates, emphasizing its criticality for safety-critical applications. The performance measure used to evaluate the results was the detection success rate. The data revealed that the spoofing validation dataset achieved a success rate of 98.77%, while the meaconing validation dataset reached a success rate of 98.72%.

VI. DL-Based Detection Methods

Analogous to the previous section, we discuss in this section various DL-based detection methods of GPS spoofing attacks that were introduced in the literature, specifically for UAVs. Table II provides an overview of the results achieved through various DL methodologies.

In [18], Galvan et al. developed a detection method using a CNN-based approach, leveraging inertial measurement unit (IMU) data within the flight control system. The authors trained the CNN model with anomalous data obtained from UAV simulators, a more cost-effective and safer alternative to using actual UAVs. Comparing this model against others, including LSTM, BiLSTM, CNN+LSTM, and ConvLSTM,

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TABLE II. SUMMARY OF RESULTS FOR DL-BASED DETECTION METHODS OF GPS SPOOFING ATTACKS ON UAVS

Ref.	AI Method	Additional Info.	Dataset	Evaluation Metrics							
				AUC (%)	*Accuracy (%)*	*Recall (%)*	*F1-Score (%)*	*Precision (%)*	*MSE*	*Detection Time*	*Detection Rate (%)*
[18]	CNN	using an accelerometer, 5 sec. time window & 20% threshold level	Developed	88.1	-	-	-	-	-	-	-
	LSTM	-		69.9	-	-	-	-	-	-	-
	BiLSTM			69.5	-	-	-	-	-	-	-
	CNN + LSTM			72.7	-	-	-	-	-	-	-
	ConvLSTM			79.3	-	-	-	-	-	-	-
[29]	Conv Autoencoder	-	UAV Attack Dataset	100	99.75	99.5	98.51	97.65	-	-	-
	LSTM Autoencoder			100	99.72	97.88	98.33	99.06	-	-	-
	Denoising Autoencoder			100	99.35	99.67	96.38	95.04	-	-	-
	OC-SVM			70.37	93.6	61.94	61.97	62.63	-	-	-
	LOF			92.47	79.49	79.77	55.43	56.12	-	-	-
	IF			95.03	98	80.08	78.96	79.18	-	-	-
[30]	CNN	-	Developed	-	-	97	99	100	-	-	-
[31]	MVSK-MLP	-	Developed	-	92	-	-	-	0.06	-	-
	BOX-MLP			-	90	-	-	-	0.075	-	-
	WD-MLP			-	93	-	-	-	0.05	-	-
[32]	CNN	-	Developed	-	88	80	86	95	-	-	-
[33]	CNN	500Hz doppler shift step and 0.5 chip	Developed	-	-	-	-	-	-	0.0648ms	>96
	GAN	500Hz doppler shift step and 0.5 chip		-	-	-	-	-	-	0.2418ms	> 95
		500Hz doppler shift step and >0.5 chip		-	-	-	-	-	-	-	> 98
	Residual Signal Detection	500Hz doppler shift step and 0.25 chip		-	-	-	-	-	-	0.0923ms	-
[34]	STL with Multiclass SVM	-	Developed	-	94	-	-	-	-	-	-
	Multiclass SVM			-	78	-	-	-	-	-	-
[35]	LSTM binary classifier	UAV-Generalized Detectors	Developed	-	97.79	97.79	97.81	97.89	-	-	-
	LSTM-Auto-encoder OCC			-	94.98	94.93	94.93	95.03	-	-	-
	LSTM binary classifier	UAV-Specific (Average		-	98.06	-	-	-	-	-	-

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	LSTM-Auto-encoder OCC	value of all UAV models)		-	97.74	-	-	-	-	-	-
[36]	ANN	-	UAV Attack Dataset	-	91.68	-	-	-	-	-	-
	Naïve Bayes			-	87.26	-	-	-	-	-	-
[37]	RNN	Highway scenario, 4 layers with 16 neurons	Developed	-	98	-	-	-	0.0031	-	-
	RNN	Smart city scenario, 3 layers with 16 neurons		-	-	-	-	-	0.0066	-	-
	LSTM-RNN	Smart city scenario, 4 layers with 32 neurons		-	98.7	-	-	-	0.0097	-	-
	GRU-RNN	Smart city scenario, 4 layers with 32 neurons		-	98.5	-	-	-	0.0097	-	-
[38]	LSTM	-	Developed	-	78	-	-	-	-	3-5s	-
[39]	LSTM	-	MAVLINK	-	99.93	85.71	92.31	100	-	-	-
			TEXTBAT	-	82.1	75.58	82.59	91.04	-	-	-
	SVM		MAVLINK	-	96	-	95.99	-	-	-	-
			TEXTBAT	-	82.3	-	83.25	-	-	-	-
	MLP		MAVLINK	-	99.93	85.71	92.29	99.96	-	-	-
			TEXTBAT	-	83.23	67.14	82.79	87.07	-	-	-
	RF		MAVLINK	-	89.33	-	89.21	-	-	-	-
			TEXTBAT	-	56.77	-	47.52	-	-	-	-
[40]	ResNet	-	Developed	-	94.8	97.9	95.4	93	-	-	-
	SqueezeNet			-	89	93.6	90.3	87.1	-	-	-
[41]	LSTM Autoencoder	-	Developed	-	-	96	-	-	-	-	-
[42]	CNN	-	Developed	-	-	-				160us	92.1
	LSTM			-	-	-				692us	91.4
	BP			-	-	-				54us	96.2
	SVM			-	-	-				-	88.2
	CBA			-	-	-				367us	99.1
	BiLSTM			-	-	-				-	90.4
	CNN-BiLSTM			-	-	-				-	92.8
	CNN-Attention			-	-	-				-	87.6
	BiLSTM-Attention			-	-	-	-	-	-	-	95.4

, they utilized the receiver operating characteristic (ROC) curve for effectiveness assessment. The CNN model, especially with accelerometer data over a 5-second window and a 20% threshold, achieved an 88.10% AUC in detecting GPS spoofing.

Fraser et al. proposed in [29] a CNN model incorporating autoencoders to identify GPS spoofing and other attacks in digital twin technology, using the UAV Attack Dataset [19] and analyzing 88 features. This model was compared with a one-class support vector machine (OC-SVM), isolation forest (IF), local outlier factor (LOF), denoising autoencoder, and LSTM autoencoder. Notably, the CNN autoencoder surpassed the LSTM variant in computational efficiency and delivered exceptional 100.00% AUC, 99.75% accuracy, 99.50% recall, 98.51% F1-score, and 97.65% precision.

Sung et al. crafted in [30] a one-dimensional CNN model using residual network (ResNet) architecture for small UAVs' spoofing detection. This model, featuring nine convolutional layers, a pooling layer, and a fully connected layer, utilizes SNR-related features. It achieved the following results: 97% recall, 99% F1-score, and 100% precision. The one-dimensional CNN's simplicity, coupled with its low computational demand and rapid detection capability, makes it a practical solution for embedding in UAVs for efficient spoofing attack detection.

In [31], three distinct MLP algorithms (WD-MLP, MVSK-MLP, BOX-MLP) were evaluated, focusing on UAV connections to varying base stations. Utilizing a real-world dataset, these models, with statistical path-loss-based features, were assessed for mean squared error (MSE) and accuracy. The WD-MLP showed superior results with over 93%

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accuracy and 0.05 MSE. Subsequently, the authors explored in [32] a CNN-based model for UAV monitoring and control, integrating mobile network assistance. This model, comprising two convolutional layers, a pooling layer, a flattened layer, and three fully connected layers, assessed path loss differences for performance evaluation. The CNN with transfer learning approach achieved 88% accuracy, 80% recall, 86% F1-score, and 95% precision, underlining the efficacy of combining CNN with transfer learning for cost-effective GPS spoofing detection in UAVs.

Li et al. proposed in [33] a method for effectively detecting spoofing signals that are highly synchronized with authentic signals using a general adversarial network (GAN). The model was trained using the authors' dataset, and its detection performance was compared to that of a CNN model and a traditional residual signal detection method. Although the GAN had a shorter training time of 1.2285ms compared to the CNN, the CNN achieved a faster detection time of 0.0751ms for a Doppler shift step of 250Hz and a pseudocode phase difference between the spoofing signal and the authentic signal 0.25 chip. However, given identical circumstances, the GAN performed slightly better than the CNN, achieving a detection probability of over 98%. Furthermore, the GAN outperformed the traditional residual signal detection method significantly.

In [34], Arthur et al. developed a cutting-edge method based on deep reinforcement learning. This method utilizes a deep-Q network specifically for dynamically adjusting the flight paths of UAVs. The technique integrates Self-Taught Learning with a multiclass SVM, targeting the precise detection of GPS spoofing, particularly in unknown areas. The architecture of the model includes an input layer, a hidden layer, and an output layer, and it processes 200 distinct attributes. The model, trained in a simulated setting, achieved a 94% accuracy in identifying GPS spoofing attacks. This capability allows UAVs to either safely return to their base or move to a secure area when such threats are detected. Although the experiment was conducted in a simulated environment, the findings suggest strong potential for applying this method in real-world scenarios.

Agyapong et al. investigated in [35] the performance of two algorithms based on LSTM for detecting UAV spoofing: one was an LSTM binary classifier, and the other was an LSTM autoencoder functioning as a one-class classifier. These algorithms underwent training with datasets created by the GAZEBO simulator and were evaluated on a range of UAV models, such as the Quadrotor and Typhoon H480, among others. The LSTM binary classifier, composed of several LSTM layers, dropout layers, a fully connected layer, and a Softmax layer, delivered good outcomes. It was particularly effective with the Quadrotor model, demonstrating a 97.79% recall rate, a 97.81% F1-score, a 97.89% precision rate, and a 99.56% accuracy in detection. Additionally, the authors conducted a comparative analysis of these models' efficiency in a simulated environment and executed hardware validation tests using the Intel NCS2.

In [36], Azaha et al. conducted a comparative analysis of the Naïve Bayes and ANN algorithms for efficient detection. They employed a publicly available dataset for this analysis. It was observed that the ANN model, composed of an input layer, a hidden layer, and an output layer, surpassed the Naïve Bayes in performance. The ANN model achieved an accuracy value of 91.68%, in contrast to the 87.26% accuracy of Naïve Bayes. Furthermore, the ANN demonstrated a reduced error rate, underscoring its enhanced effectiveness in identifying GPS spoofing attacks.

Xiao et al. delved in [37] into the effectiveness of various recurrent neural network (RNN) models for GPS spoofing detection. They focused on different environments, including smart city and highway scenarios. The models evaluated included a basic RNN, a long short-term memory recurrent neural network (LSTM-RNN), and a gated recurrent unit recurrent neural network (GRU-RNN). It was found that each model's performance was optimized for specific scenarios. In the highway environment, the basic RNN model emerged as the most effective, whereas the LSTM-RNN showed superior performance in the smart city setting. The LSTM-RNN, designed with four layers and 32 neurons, registered an accuracy of 98.70% and an MSE of 0.0097 in the smart city scenario. In contrast, the basic RNN model, equipped with 16 neurons, attained an accuracy of 98% and an MSE of 0.0031 in the highway context.

Wang et al. introduced in [38] a detection method that integrates an LSTM neural network with a Kalman filter, setting it apart from conventional methods that solely rely on the Kalman filter. This algorithm, developed through real-world experiments, incorporates many features, including velocity and GPS coordinates. Its performance was assessed based on two key metrics: accuracy and detection speed. The algorithm achieved an accuracy of 78% and could detect issues within 3-5 seconds. However, a quicker detection time might be necessary for effective prevention of spoofing attacks.

Jullian et al. embarked in [39] on a comparative analysis of Multilayer Perceptron (MLP) and LSTM models in their capacity to detect GPS spoofing attacks. They trained and evaluated these models on two distinct datasets, MAVLINK and TEXTBAT, utilizing features like GPS coordinates and position logs. The configuration of the MLP model included four hidden layers for MAVLINK and three for TEXTBAT. In contrast, the LSTM model was structured with three hidden layers for MAVLINK and two for TEXTBAT. The study revealed that the MLP model was more effective on the TEXTBAT dataset, achieving an accuracy of 82.23%, a precision of 87.07%, a recall rate of 67.14%, and an F1-score of 82.79%, thus surpassing the LSTM model. The MLP model's performance was further validated through comparative analyses with other algorithms, including Random Forest, SVM, and autoencoder.

Xue et al. introduced in [40] DeepSIM, which employs ResNet for on-ground detection, and Squeeze Network (SqueezeNet) for on-board detection to compare the real-time location of the UAV with a satellite image of the same area. The authors also employed visual features from their SATUAV dataset in their model and used data augmentation techniques to improve the collected features. The authors concluded that the ResNet approach achieves high results with

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94.8% accuracy, 93.0% precision, 97.9% recall, 5.2% error rate, and 95.4% F1-score, which is far better than the SqueezeNet approach. The major disadvantage of DeepSIM is that it relies heavily on satellite imagery, which can be expensive and may only sometimes be available in real-time scenarios.

Tlili et al. proposed in [41] an LSTM autoencoder using the UAV Attack Dataset [19]. The proposed approach achieved a high detection accuracy of 96%. However, a limitation of this paper is the need for more comparison with other existing methods, making it difficult to assess the effectiveness of the proposed approach compared to other state-of-the-art methods.

In [42], Wu et al. introduced a novel technique for real-time detection of drone spoofing attacks through the analysis of sensor data. They developed a sophisticated model called convolutional neural network (CNN) – bidirectional long short-term memory (BiLSTM) -Attention, also known as CBA, which was specifically designed to differentiate between DoS attacks and GPS spoofing accurately. This model surpassed various other classification methods, such as CNN, long short-term memory (LSTM), back-propagation, and SVM. An essential aspect of the proposed approach was the incorporation of the Shapley additive explanations (SHAP) technique, which greatly enhanced the transparency of the model's decision-making process, enabling a clearer understanding of how the model reached its conclusions. The effectiveness of this system was validated in scenarios involving both spoofing and DoS attacks, showcasing its versatility in combating different types of threats. The authors also emphasized the importance of carefully selecting UAV sensor data, particularly GPS data with specific characteristics. This careful selection helps optimize computational efficiency and ensures rapid detection of attacks.

Whelan et al. compared in [43] three different algorithms, OC-SVM, LOF, and Autoencoder, to detect GPS spoofing as a typical example of an external sensor-based attack. They applied their algorithms on six different types of UAVs: 3DR IRIS+, Holybro S500, Yuneec H480, DeltaQuad VTOL, Standard Tailsitter, and Standard Plane. The obtained results show that the DeltaQuad VTOL using the autoencoder algorithm achieved the highest F1-score, 99.73%, while the 3DR IRIS+, Holybro S500, and Yuneec H480 using the LOF algorithm achieved the highest recall value: 100%. Furthermore, DeltaQuad VTOL and Standard Tailsitter, using the autoencoder algorithm, achieved the highest precision value: 87.728%. The authors considered the F1-score to determine the overall highest performer. Based on the findings, it was concluded that the autoencoder is the preferred algorithm for this approach, with an average F1-score of 94.81%. The OC-SVM and LOF algorithms showed average F1-scores of 81.17% and 58.93%, respectively.

Park et al. proposed in [44] an IDS that utilizes an autoencoder, a deep neural network (DNN) of unsupervised learning, to detect GPS spoofing attacks on UAVs. The authors trained the model using only benign data from the UAV attack dataset [19] and considered the following features: location, position and orientation, IMU, and system status. The proposed model identifies an attack by detecting significant differences between reconstruction losses from the benign flight and the flight under attack. To prevent overfitting, the authors utilized L1 and L2 regularizes. They also applied batch normalization to the encoder and decoder and optimized parameters with Adam optimizer to ensure practical model training. The authors concluded that the proposed model requires minimal data labeling effort compared to ML-based models.

Ahn et al. introduced in [45] an unsupervised one-dimensional CNN model for detecting anomalies in UAVs using INS data. The dataset used in the paper consisted of actual INS data from x70 drones. To improve the model's performance, a PCA feature extraction method was employed, and the model architecture consisted of 6 hidden layers. The following features are considered in their analysis: Inertial Position and Velocity, Accelerometer, Gyroscope, Position, Navigation State, and Arming State. The authors highlighted the limitations of using a two-dimensional CNN model for INS data due to the lack of spatial relationships between columns of the used algorithm. A limitation of the work was the absence of a comparison between the proposed model and other DL models, as well as the lack of providing results to analyze the performance of the proposed work.

VII. Discussion

Most of the addressed methods performed simulations to validate their proposed models, and hence, they may not fully capture the complexities and challenges of real-world scenarios. Most of the authors also developed their datasets, while few of them used a common dataset, such as the open-source UAV Attack Dataset [19], as shown in Table I and Table II. Some datasets are more significant than others, and it is well known that the effectiveness of an AI-based algorithm depends heavily on the dataset.

Moreover, the adopted performance metrics to evaluate the proposed models vary from one work to another, as discussed in the paper and are shown in Table I and Table II. It is worth mentioning that the choice of metric should depend on the specific operational scenario of UAV use. For example, in military applications where missing an attack could be catastrophic, recall and detection rate might be prioritized. In contrast, in commercial delivery services, where false alarms could disrupt operations and cause economic losses, precision might be more important. A balanced approach can also be adopted using a combination of metrics that provide a holistic view of system performance to accommodate various operational scenarios. Therefore, when providing a new detection method, it is important for the researcher to highlight how metrics they adopt reflect the operational priorities and risk management strategies in their work.

It is also well-known that the complexity of AI-based algorithms and their computational load usually play a significant role in their performance. Most of the works should have reported the computational complexity of the proposed algorithms and their effect on the performance of the UAVs in real operational scenarios. Increasing the complexity of an algorithm to achieve better GPS spoofing attack detection

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capabilities may not be suitable to be used in UAVs, taking into consideration that the operation of UAVs faces many realistic constraints, such as the computational capabilities of the embedded system's processor, the power consumption, and the on-board battery size, among others.

Although many kinds of onboard motion sensors and vision sensors are used in the navigation systems of UAVs [10]-[11], the data they provide could have been more efficiently utilized in developing the detection methods of GPS spoofing attacks on UAVs in the literature. Many multi-sensor fusion detection methods for such attacks are introduced in the literature for Autonomous Vehicles. As UAVs are a type of Autonomous Vehicle but with more special operational constraints than the other types, these detection methods can be adopted and modified to suit the UAVs' operational characteristics.

VIII. Conclusion

In conclusion, GPS spoofing attacks are significant threats to UAVs. Detecting and preventing such attacks is essential for their safety, security, and proper operation. In this paper, we provided an overview of the different types and levels of GPS spoofing attacks on UAVs, in addition to the AI-based methods that have been introduced in the literature to detect such attacks.

We showed that DL and ML are promising approaches for detecting GPS spoofing attacks on UAVs, highlighting the reported results of the many works in the literature. AI-based algorithms can learn to recognize patterns and anomalies in GPS signals that are indicative of spoofing attacks and can provide real-time detection and prevention of such attacks. However, there are still many challenges that need to be addressed to make AI-based detection methods more effective and reliable, as discussed in Section VII. In summary, the need for a large and common dataset built from multiple scenarios to conclude which DL and ML algorithm is the best for detecting such attacks; the complexity of the used AI-based algorithms, especially for the DL-based detection methods, and their impact on the UAVs' resources; the reflection of the adopted metrics on the operational priorities and risk management strategies; and the use of on-board sensors, both inertial and vision sensors, more effectively. Furthermore, while the addressed papers' results show promise for UAV-specific detection algorithms, it is essential to note that many of the validations were conducted in a simulated environment and not tested in real-world scenarios, which could limit the generalizability of the findings or their accuracy.

An essential recommendation for researchers and developers is to adopt a common dataset and similar performance metrics to evaluate an AI-based algorithm when developing new GPS spoofing detection methods on UAVs. This will reflect the level of enhancement of the proposed method when compared to others.

Overall, AI-based methods have great potential for enhancing the security and safety of UAVs against GPS spoofing attacks. Our survey provides a valuable resource for researchers working in this field. It inspires future work to introduce more advanced and effective methods for detecting GPS spoofing attacks on UAVs. It is also helpful for practitioners to deploy effective UAV security systems based on the summarized findings of the conducted research in the literature.

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